

# Computing Consistent Levels for Layered Software Architecture Visualizations

## *Separating Architectural Hypothesis, Package Tree and Local Layout*

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**Keywords:** software architecture visualization, layout invariants, layered architecture visualization, strongly connected components, dependency analysis, software engineering tools

**Abstract:** Layered architecture visualizations communicate dependency structure through vertical position: elements that depend on others sit higher, foundational elements sit lower. Their value rests on a simple promise—an upward edge should reveal an architectural anomaly, not a layout artifact. If level computation is wrong, the drawing still looks plausible while misleading the viewer.

Levelized dependency maps are not new. Structure101’s Levelized Structure Map arranges elements into rows and exposes feedback in cyclic regions (Headway Software Technologies Ltd., 2023b; Muccini and Tekinerdogan, 2012; Headway Software Technologies Ltd., 2023a). Published descriptions specify the intended visual behaviour but not the concrete level-computation algorithm or its consistency conditions, leaving it unclear whether an upward edge is a finding, a tolerated cycle cut, or an implementation defect.

In this paper, we present the level pipeline implemented in S202, an open-source Java architecture-visualization tool under Apache 2.0. The pipeline has three stages on three independent graphs. First, S202 detects global SCCs in the class graph and heuristically cuts large cycles. Second, from weighted package dependencies it derives an architectural hypothesis: more heavily used packages sit lower, their users higher. Third, it computes a local level inside each parent container, which determines the vertical placement of siblings.

The algorithmic contribution is the strict separation of architectural hypothesis from layout position. Package analysis decides which package directions are normal, cyclic, or heuristically cut; local placement only decides where classes and visible containers appear inside one concrete parent. This prevents feedback loops in which class feedback, package order, and local positions redefine each other.

Five machine-checkable invariants verify that each run is internally consistent: class dependencies respect class levels after heuristic cuts, dominant weighted package dependencies respect package levels, cyclic peer packages share a level after filtering, visual ranks follow local levels, and cached edge classifications match the current SCC and level state. The checks distinguish pipeline defects from deliberately tolerated heuristic cuts and from architecture violations that the visualization is meant to expose.

We demonstrate the pipeline on Minecraft Forge 1.19.2, a large and highly cyclic system that stresses every stage. Separating global architecture depth from local layout, and preserving the computed package order in the UI instead of reordering it to suppress cycles, reduces the visual upward-edge count from roughly 6000 to 4200; the remaining edges are real architecture violations or explicit heuristic cuts. The result is a corrected, reproducible level-computation algorithm whose visible anomalies can be explained and checked.

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## Appendix A – Why Naive Single-Pass Does Not Work: Failure Modes

### The Core Problem: Architectural Hypothesis, Local Layout, Five Consistency Checks

A correct layered layout has to separate an architectural hypothesis from the local placement that makes this hypothesis visible. In S202 this means answering three related questions that are often treated as one ordering problem:

- *Class analysis* – where does a class sit in the global class dependency stack? This is the longest-path depth in the SCC-collapsed class graph.
- *Package analysis* – what package order is implied by the aggregated dependency traffic? This architectural hypothesis is computed on the weighted inter-package graph, independently of the deepest contained class.
- *Local layout* – where should an element’s box sit inside its parent container? Only dependencies between classes in the subtrees of that parent’s direct children decide this local level.

The three computed values are independent in principle, and the implementation must keep the package-level hypothesis separate from local layout choices. Mixing them is the root cause of every failure mode below. Five consistency checks then inspect the resulting levels and rendered order:

1. **Class-level consistency (R1-algo):** If  $A \rightarrow B$  denotes a class dependency, then  $\text{archLvl}(A) > \text{archLvl}(B)$  after heuristic back-edges are removed. This checks the longest-path assignment in the class-dependency SCC-DAG.
2. **Visual-rank consistency (R1-visual):** For every class dependency  $A \rightarrow B$  that is not a heuristic back-edge, the rendered position of  $A$  in its parent box chain sits above  $B$ . The comparison is lexicographic on the tuple of local levels along the parent path, ending with the class’s own local level. This is the statement represented by the drawing.
3. **Package-SCC consistency (R2):** Packages that form a cyclic group in the filtered package dependency graph share the same package architecture level.
4. **Package-order consistency (R3):** In the weighted inter-package dependency graph, when the call traffic from package  $P_A$  to  $P_B$  dominates the reverse,  $P_A$  ends up at a strictly higher package architecture level than  $P_B$ .
5. **Classification consistency (R5):** Cached edge classifications such as `NORMAL`, `VIOLATION`, and `INTRA_SCC` match the current level assignment and SCC membership.

R1-algo, R2, R3 and R5 check the algorithmic model; R1-visual checks the rendered order. The failure modes that follow show why each obvious shortcut collapses one of these concerns into another and loses information.

### Relation to Structure101 and Levelized Structure Maps

Structure101’s Levelized Structure Maps are a closely related practical predecessor to the layout studied here. They organize code elements into rows such that, as far as possible, dependencies flow downward, and they expose feedback dependencies when cyclic regions prevent strict levelization (Headway Software Technologies Ltd., 2023b; Headway Software Technologies Ltd., 2023a). This makes Structure101 an important point of comparison and a necessary part of the related work.

The public Structure101 documentation, however, describes the intended properties of the visualization rather than the concrete algorithm that computes the rows. The same is true for academic references that discuss Structure101 as a software architecture analysis tool or as related work for layered visualization (Muccini and Tekinerdogan, 2012): they characterize the LSM concept, but do not provide an algorithmic specification of Structure101’s levelization procedure. To the best of our knowledge, no published paper describes the exact Structure101 algorithm.

S202 therefore provides an open algorithmic account of a related class of levelized dependency layouts, not a claim that the visual idiom itself is new. The novelty lies in making the computation explicit, separating class architecture depth, package architecture order, and local layout position, and checking the resulting layout through executable invariants.

## Failure Mode 1: Cascading Level Updates

An algorithm iterates over classes and assigns levels. It first sees class  $A$  in package  $P_1$ , computes  $\text{level}(A) = 3$ , and therefore sets  $\text{level}(P_1) = 3$ . Later it reaches class  $B$  in package  $P_2$  and discovers that  $A$  and  $B$  are in the same SCC and must share a level, now  $\text{level}(A) = \text{level}(B) = 5$ . This forces  $P_1$  to be recomputed. If  $P_1$  depends on  $P_3$ ,  $P_3$  may need to move as well. A cascade follows.

In a naive single-pass implementation, however,  $P_1$  and  $P_3$  may already have been finalized. SCC membership is only known after the relevant graph has been traversed; if level assignment and SCC discovery are interleaved, the iteration order decides whether the correction is still applied in time.

## Failure Mode 2: Interleaved Cycle Detection Produces Classpath-Dependent Results

The most subtle practical failure mode occurs when cycle detection and level computation are not separated. The algorithm traverses the class graph, assigns levels, and handles discovered cycles on the fly, depending on which class happens to be processed first.

Classes come from JARs. The order in which the JVM sees classes follows the order in which they were packaged into the JAR. That order can depend on the build tool, module order in a multi-module build, filesystem ordering on the build server, and other implementation details. None of these orders is semantically meaningful; none encodes architectural dependency direction.

The result is unreliable: the same source code can produce different level layouts under different build configurations. The errors are subtle. The layout may still look plausible, arrows may mostly point in the expected direction, and no obvious visual break appears. Yet the precise hierarchy information is corrupted and may remain unnoticed for a long time.

The solution is to separate the phases strictly: first compute the SCC partition on the complete graph, then build the SCC-DAG, then assign levels. Tarjan's algorithm computes the SCC partition in  $O(V + E)$ ; with deterministic input ordering, the implementation becomes reproducible and independent of classpath traversal artifacts.

## Formal Counterexample: Why a Naive Recursive Algorithm Fails

The intuitive level-computation algorithm is a recursive DFS with memoization:

```
level(A) :
  if A already computed: return level(A)
  if A has no dependencies: return 0
  return max(level(dep) for dep in A.dependencies) + 1
```

For acyclic graphs this is correct and equivalent to longest-path computation on a DAG. For cyclic graphs it fails structurally.

### Counterexample with two SCCs:

```
SCC_1 = {A, B}, with A -> B -> A
SCC_2 = {C, D}, with C -> D -> C
Cross dependency: A -> C
```

The correct levels are  $\text{level}(C) = \text{level}(D) = 0$  and  $\text{level}(A) = \text{level}(B) = 1$ .

Recursive algorithm, starting at  $A$ :

```
level(A)  [A "in progress", sentinel = 0]
-> level(B)
    -> level(A): cycle detected, return 0
    B.level = 0 + 1 = 1
-> level(C)  [C "in progress", sentinel = 0]
    -> level(D)
        -> level(C): cycle detected, return 0
        D.level = 0 + 1 = 1
    C.level = 1 + 1 = 2    <-- wrong; correct would be 0
    A.level = max(1, 2) + 1 = 3    <-- wrong; correct would be 1
```

After SCC equalization,  $SCC_1 = \max(3, 1) = 3$  and  $SCC_2 = \max(2, 1) = 2$ . Both results are wrong.

The cause is that the recursive algorithm enters  $SCC_2$  from the context of  $SCC_1$ . The sentinel-based local cycle handling inflates the level of  $SCC_2$ , and that inflated level then feeds into the computation of  $SCC_1$ . The correct level of  $SCC_2$ , namely zero, is only known after the graph has been collapsed into SCCs and evaluated as an SCC-DAG.

Thus, recursive level algorithms that mix traversal, cycle handling, and level assignment can produce traversal-order-dependent results on graphs with interacting SCCs. In JVM-based tools, that traversal order can be influenced by classpath and packaging order. The correct solution collapses SCCs before level assignment: Tarjan, then SCC-DAG, then longest path.

### Failure Mode 3: Deriving Package Levels from Class Levels

An obvious implementation assigns each package a level equal to the maximum level of its contained classes, then lifts parent packages transitively. This is wrong for two reasons.

First, the rule conflates containment with dependency. A package that *contains* a class at level 7 is not necessarily architecturally positioned at level 7 itself; it may have no cross-package dependencies at all. Two disconnected package trees processed from the same JAR would be ordered relative to each other by accident of their class contents, not by their actual dependency relationships.

Second, the approach creates a circular dependency between the two level systems: class levels depend on the class graph, but package levels derived from class levels then constrain the visual position of packages relative to classes, making the combined coordinate system ill-defined.

S202 avoids this by computing class levels and package levels on independent graphs with no mutual constraint. Each system starts at zero and grows according to its own dependency structure.

### Failure Mode 4: Large SCCs Without Heuristics Collapse the Layout

Without a mechanism for breaking large SCCs, all members of a cyclic group must be placed on the same level. For small cycles of two to five classes this is acceptable. For Minecraft Forge 1.19.2 (5 123 classes), however, we observed a single SCC containing 2 857 classes. Without a breaker, more than 55 percent of the project would be assigned the same level. All remaining classes would be distributed above, but the internal dependency hierarchy within this large cycle would be invisible.

The heuristic SCC breaker solves this practical problem. It identifies back edges—dependencies that run against the inferred architecture direction—and removes them for level computation. The removed edges are visualized as violations instead of being ignored. The layout preserves visibility of cyclic dependencies while still producing a usable hierarchy.

### Relation to Feedback Arc Set Heuristics

Failure Mode 4 is an instance of the Minimum Feedback Arc Set problem: remove a minimal set of directed edges so the graph becomes acyclic. The problem is NP-hard (Karp, 1972), so any practical polynomial-time implementation must use a heuristic or approximation.

A well-known related heuristic is due to Eades, Lin, and Smyth (Eades et al., 1993). It orders nodes according to the difference between outgoing and incoming degree, then classifies edges that point backward in the resulting sequence as feedback edges. The intuition is similar to S202: high out-degree suggests a higher-level element; high in-degree suggests a lower-level element.

S202 uses a deliberately simpler variant: a direct normalized score per node,

$$r(P) = \frac{\text{outWeight}(P) - \text{inWeight}(P)}{\max(1, \text{outWeight}(P) + \text{inWeight}(P))},$$

where the weights are summed within the SCC under consideration. At the class level the weights are the in- and out-degrees in the class dependency graph; at the package level they are aggregated method-call counts. An edge  $P_A \rightarrow P_B$  inside an SCC is classified as a back-edge candidate when  $r(P_A) < r(P_B) - 0.1$ . Equal-rank pairs represent symmetric bidirectional dependencies with no dominant architectural direction; they remain in the same SCC and are assigned the same level.

The important distinction is transparency. Many tools break cycles implicitly, but S202 classifies heuristic cut edges explicitly. They are marked as `VIOLATION`, rendered in red, and known to the invariant checker. This makes algorithmic defects separable from tolerated heuristic artifacts.

## The Solution: Three Computed Values on Independent Graphs

Failure modes 1 to 3 share a root cause: conflating values that belong to different semantic domains. S202 resolves this by computing three independent values, two of them for analysis and one for layout.

- **Class architecture level** is the longest path in the SCC-collapsed class dependency DAG. It represents global architectural depth: how many layers of dependencies separate a class from the bottom of the analysed dependency graph.
- **Package architecture level** is the longest path in the SCC-collapsed weighted inter-package dependency graph. Edge weights are method-call counts aggregated from class-level references. The result represents the module's position in the system's overall dependency order, independently of the class-level coordinate.
- **Local level** is the longest path in the SCC-collapsed sibling-only dependency graph of a single parent container. The graph contains only edges whose source and target both lie in some direct child of that parent; references leaving the parent's subtree are excluded because they are handled by the enclosing container. Every element (class and package) gets a local level that is only meaningful within its own parent box. The renderer sorts each parent's children by this index, so the visible Y position of a box is the local level, not the global architecture level.

All three concepts use the same algorithmic shape—Tarjan SCC detection, weight-based back-edge identification, DAG condensation, longest-path propagation—but on three different graphs with three different scopes. Because the systems are independent, packages with no cross-package dependencies correctly land at architecture level 0 regardless of the class levels inside them, and a top-level class like *ArchitectureView* that orchestrates a sub-package such as *ui.rendering* sits visually above the sub-package box without distorting either of their architecture levels.

The separation is an engineering choice driven by semantic clarity. A unified single-coordinate approach is theoretically possible, but it conflates three semantically distinct concerns—global architectural depth, module dependency order, and per-container visual position—into one number, making the pipeline hard to reason about, test, and maintain. Each simplification of such a unified approach turns into one of the failure modes above.

The three concepts have the following algorithmic structure. Phases 1 and 2 compute the global architecture levels; Phase 3 derives the local level per parent container from the architecture-level output.

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Algorithm 1: Phase 1 – Global Class Levels (`LevelCalculator`, Step 3)

---

```

class dependency graph
  → Tarjan (SCCs)
  → SCCBreaker: identify heuristic back-edges in large SCCs ( $|S| \geq 3$ ; 2-class SCCs are equalized instead)
     $\text{rank}(P) = (\text{outDeg} - \text{inDeg}) / \max(1, \text{outDeg} + \text{inDeg})$ 
    edge  $A \rightarrow B$  is a back-edge candidate if  $\text{rank}(A) < \text{rank}(B) - 0.1$ 
  → Remove back-edges → filtered class DAG
  → Condense into SCC-DAG → longest-path pass
  → Assign SCC level to all member classes (equalization implicit)

```

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**Algorithm 2: Phase 2 – Package Architecture Levels (LevelCalculator, Step 4)**

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**Build weighted inter-package dependency graph  $G_P = (V_P, E_P, w)$ :**

For each class  $C$  in package  $P_{\text{leaf}}$  with method calls to class  $D$  in  $P_{\text{target}}$  ( $P_{\text{leaf}} \neq P_{\text{target}}$ ):

$w(P_{\text{ancestor}}, P_{\text{target}}) += \text{callCount}(C \rightarrow D)$

for every ancestor  $P_{\text{ancestor}}$  of  $P_{\text{leaf}}$  with  $P_{\text{ancestor}} \neq P_{\text{target}}$

Fallback weight = 1 for structural references with no method-call data

No directional filtering: both parent→child and child→ancestor edges enter  $G_P$ . The SCC breaker decides direction by weight.

**Iterative SCC-breaking (repeat until stable):**

Tarjan on current graph → find SCCs with  $|S| \geq 2$  (2-package SCCs broken if rank differs; equalized otherwise)

For each member  $P \in S$ :

$$r(P) = (\sum_{Q \in S} w(P, Q) - \sum_{Q \in S} w(Q, P)) / \max(1, \sum_{Q \in S} w(P, Q) + \sum_{Q \in S} w(Q, P))$$

Edge  $P_A \rightarrow P_B$  inside  $S$  is a back-edge if  $r(P_A) < r(P_B) - 0.1$ ; remove it

Equal-rank pairs remain in the SCC → assigned the same level

**Level assignment:**

Condense remaining graph into SCC-DAG

Longest-path propagation on SCC-DAG

Assign SCC level to all member packages

---

Phase 1 produces classpath-independent class architecture levels because SCCs are collapsed before levels are assigned. Phase 2 produces package architecture levels directly from package dependencies: every class reference contributes to the weighted graph regardless of the containment relation between source and target package, and a per-SCC rank score decides which direction of a cyclic group dominates. The two architecture coordinate systems do not constrain each other—a package with no outgoing cross-package calls lands at level 0 regardless of the class levels inside it, and conversely the class levels of a package’s contents do not lift the package itself.

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**Algorithm 3: Phase 3 – Local Level per Parent Container (LocalLevelCalculator, Step 6)**

---

**for each** parent package  $P$  with at least two direct children:

Let  $S(P)$  be the direct children (classes and sub-packages).

Build sibling-only weighted graph  $G_P^{\text{loc}}$  on  $S(P)$ :

edge  $X \rightarrow Y$  with weight  $\sum \text{callCount}(c \rightarrow d)$

over all class refs  $c \rightarrow d$  where  $c$  lies in the subtree of sibling  $X$

and  $d$  lies in the subtree of sibling  $Y$ ;

refs leaving the parent’s subtree are excluded.

Tarjan on  $G_P^{\text{loc}} \rightarrow \text{SCCs}$ .

Within each SCC of size  $\geq 2$ : evaluate every candidate edge cut.

First cut an edge that contradicts the package architecture; only then

prefer the cut that maximally decomposes the SCC; edge weight and rank as further tie-breakers.

Condense to SCC-DAG, longest-path pass, assign **localLevel** per sibling.

---

Phase 3 is a pure layout phase. It never modifies architecture levels. Each parent container gets its own computation: a class’s local level inside *ui* carries no relationship to a class’s local level inside *ui.rendering*. The renderer sorts each parent’s children by their local level, descending. The enclosing parent’s local level handles cross-container ordering recursively. The local SCC breaker resolves local cycles—for example, two sibling sub-packages with mutual class references—silently. It removes one edge at a time, always preferring an edge that contradicts the package architecture: Phase 3 may never reorder packages relative to the order computed in Phase 2. Only when no architecture-contradicting edge is available does the breaker fall back on cycle-decomposition score, edge weight, and rank. S202 does not surface these local cycles as a separate Tangle category because the user-visible tangle list remains a global architecture statement.

|                    | Phase 1 – Class     | Phase 2 – Package              | Phase 3 – Local Level              |
|--------------------|---------------------|--------------------------------|------------------------------------|
| <b>Graph</b>       | Class dependency    | Weighted inter-package         | Sibling-only weighted (per parent) |
| <b>Edge weight</b> | Unweighted          | Method-call counts             | Method-call counts                 |
| <b>Scope</b>       | Global              | Global                         | Per parent container               |
| <b>Meaning</b>     | Class arch. depth   | Module dep. position           | Visual Y in parent box             |
| <b>Zero point</b>  | Class with no deps  | Package with no outgoing calls | Bottom sibling in box              |
| <b>Cycles</b>      | SCCBreaker (degree) | Weight-rank, all edges         | Max-cycle-break within siblings    |
| <b>Used by</b>     | R1-algo, Tangles    | R3, R2                         | R1-visual, renderer sort           |

Table 1: The three independent values computed by S202.

## Layout Invariants for Level Correctness

The three level phases are guarded by five machine-checkable consistency checks. The `LayoutInvariantChecker` checks R1-algo, R2, R3, and R5 as algorithm consistency checks after every analysis run; the architecture builder checks R1-visual through violation detection on the visual-rank tuple. These checks are evidence that the computed model is internally consistent; they are not the primary algorithmic contribution.

| Rule      | Consistency check                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1-algo   | Non-heuristic class dependencies respect class architecture levels: $A \rightarrow B \Rightarrow \text{archLvl}(A) > \text{archLvl}(B)$ . Tests the longest-path assignment of Phase 1.                                                                                                                                                                                                                      |
| R1-visual | For every non-heuristic class dependency $A \rightarrow B$ , the rendered position of $A$ sits above $B$ . The check compares positions lexicographically on $[L(\text{ancestor}_1), \dots, L(\text{ancestor}_n), L(A)]$ versus $[L(\text{ancestor}'_1), \dots, L(\text{ancestor}'_m), L(B)]$ where $L$ denotes the local level. Covers same-parent, same-grandparent, and cross-tree comparisons uniformly. |
| R2        | Packages forming a cyclic group in the filtered package dependency graph share the same package architecture level.                                                                                                                                                                                                                                                                                          |
| R3        | For every weighted package edge $(P_A, P_B)$ with $w(P_A, P_B) > w(P_B, P_A)$ that is not a back-edge cut by the SCC breaker: $\text{archLvl}(P_A) > \text{archLvl}(P_B)$ .                                                                                                                                                                                                                                  |
| R5        | Cached edge classifications (NORMAL, VIOLATION, INTRA_SCC) remain consistent with the current level assignment and SCC membership.                                                                                                                                                                                                                                                                           |

Table 2: Five layout consistency checks verified after every analysis run. R1-algo, R3 and R2 check the algorithm; R1-visual checks the rendered picture; R5 keeps the cached annotations consistent.

Throughout the iterative development of the level pipeline, R2 surfaced implementation bugs after algorithm changes that the unit-test suite had not detected. Figure 1 shows a representative R2 report. In *software-ekg-7*, for instance, an R2 finding exposed a missing package-SCC equalization step and directly triggered a pipeline fix.



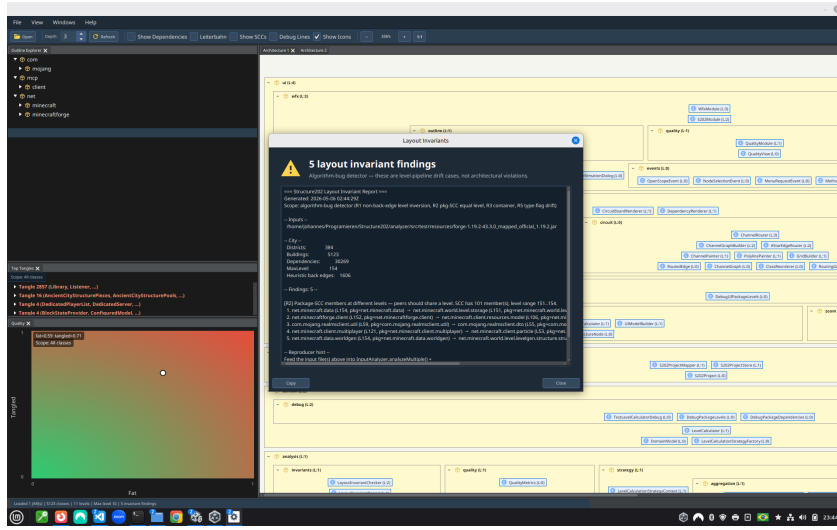


Figure 1: R2 checker report: packages in a filtered package-SCC occupy different levels. The checker reports findings as algorithm-bug candidates with a reproducer block, which makes the visual failure traceable to concrete level-pipeline consistency conditions.

R2 also guided refinement of the rule itself. Against Minecraft Forge (2 895 classes, 220 packages, 188 of 207 packages in a single package-level SCC), two R2 reports turned out to be *false positives*: R2 filtered class-level heuristic back-edges but not the package-level back-edges that the package SCC-breaker had already removed. R3 already applied the correct filter; once R2 was aligned, the Minecraft findings disappeared and all 155 invariant tests pass. The invariant checker connects the visual symptom to a reproducible algorithm defect.

## The Remaining Rules: R1-algo, R1-visual, R3, and R5

**R1-algo** ( $A \rightarrow B \Rightarrow \text{archLvl}(A) > \text{archLvl}(B)$  on the filtered class graph) is the most direct invariant on the analysis side: it is a corollary of longest-path computation on the SCC-DAG and holds trivially for any correctly implemented class-level assignment. Its value lies in providing an immediate check—a failing R1-algo signals a fundamental breakdown in the class-level pipeline.

**R1-visual** (lexicographic comparison of local-level tuples along the parent chain) is the invariant that the drawing represents. We separate the two checks intentionally: R1-algo asks “did Phase 1 compute consistent class architecture levels?”, R1-visual asks “does the rendered Y position reflect the class dependencies?” Both pass on a correct implementation; only the second one may fail when the sibling-only graph of some parent exposes a genuine upward edge, because that is a real architecture violation rather than an algorithm bug. The single visual-rank comparison covers every pair: siblings inside the same container compare on their final tuple element; cross-tree pairs diverge earlier; classes versus sub-package boxes in the same parent compare on the shared parent context. The comparison needs no special-case logic for parent→child versus sibling versus cross-package class references.

**R3** verifies that the dominant direction in the weighted package dependency graph appears in the architecture level assignment. For any package edge  $(P_A, P_B)$  where  $w(P_A, P_B) > w(P_B, P_A)$ —meaning  $P_A$  calls into  $P_B$  significantly more than vice versa—the invariant requires  $\text{archLvl}(P_A) > \text{archLvl}(P_B)$ . R3 excludes package back-edges identified during SCC-breaking from this check, analogously to how R1-algo excludes class-level heuristic back-edges. A failing R3 indicates that the weight-based SCC-breaking cut the wrong package edge, or that the package-level DAG propagation did not converge correctly.

**R5** validates that every edge’s stored classification (NORMAL, VIOLATION, INTRA\_SCC) is still consistent with the current level assignment and SCC membership. The analysis computes edge labels and caches them in the model; after any algorithmic change they can silently become stale. R5 is therefore a cross-cutting consistency check: it ties the structural output of the level pipeline back to the semantic annotations visible to the user.

## R4: A Retired Rule and What It Teaches

An early version of the invariant checker contained R4:

`Cyclic child packages must share a level.`

The rule sounds reasonable and is correct in simple cases. In practice it failed because the raw package graph contains relationships that should not be interpreted as peer-level architectural cycles:

- **Parent-child edges** create apparent bidirectional relationships because a parent contains its children and children belong to their parent. Without filtering, many parent/child pairs look cyclic.
- **SCCBreaker back edges** connect packages that are not necessarily architectural peers. A rule that includes these edges equalizes the wrong candidates.
- **Shared class SCCs** can make two packages appear package-dependent without representing a true structural peer relationship.

R4 was replaced by R2, which expresses the same intention on the correctly filtered graph. R2 removes class-level heuristic back-edges, package-level SCC-breaking back-edges, intra-subtree edges, and edges caused by shared class SCCs before running Tarjan on the remaining package graph.

Specifying invariants for a three-value level pipeline turns out to be harder than it appears. The checker therefore provides a second safety net: it not only tests the implementation against the rules, but also exposes when the rules themselves are too broad or too coarse. False positives force the rule to become more precise. The later R2 finding in *software-ekg-7* shows that the refined rule was precise enough to distinguish real pipeline bugs from apparent ones.

## Implementation: The Complete Pipeline

The conceptual structure expands into the following concrete implementation steps, which constitute the reference implementation (Appendix B).

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Algorithm 1: Complete Level Pipeline (LevelCalculator)

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- Step 1:** Create class objects (all levels  $\leftarrow 0$ )
- Step 2:** Create package objects (all levels  $\leftarrow 0$ )
- Step 3:** Compute class architecture levels (Phase 1)
- Tarjan on class dependency graph
  - SCCBreaker: identify heuristic back-edges in large SCCs ( $|S| \geq 3$ ):
    - $r(C) = (\text{outDeg} - \text{inDeg}) / \max(1, \text{outDeg} + \text{inDeg})$ ;
    - edge  $A \rightarrow B$  is a back-edge candidate if  $r(A) < r(B) - 0.1$
  - Remove identified back-edges → filtered class DAG
  - Condense into SCC super-nodes
  - Single longest-path pass on condensed DAG
  - Assign SCC level to all member classes
- Step 4:** Compute package architecture levels (Phase 2)
- 4a.** Build weighted inter-package dependency graph:
    - $w(P_A, P_B) = \sum \text{method-call counts from subtree}(P_A) \text{ to } P_B$ ;
    - propagate weights up to all ancestor packages (no directional filtering);
    - both parent→child and child→ancestor edges enter the graph
  - 4b.** Iterative package SCC-breaking (repeat until stable):
    - Tarjan on current graph → SCCs with  $|S| \geq 2$ ;
    - $r(P) = (\sum_{Q \in S} w(P, Q) - \sum_{Q \in S} w(Q, P)) / \max(1, \dots)$ ;
    - edge  $P_A \rightarrow P_B$  is a back-edge if  $r(P_A) < r(P_B) - 0.1$ ; remove it;
    - equal-rank pairs remain in the same SCC → assigned the same level
  - 4c.** Level assignment on condensed package SCC-DAG:
    - longest-path propagation; assign SCC level to all member packages.
    - No child–parent lift: layout positioning is based on `localLevel`.
- Step 5:** Set reverse dependencies (dependents)
- Step 6:** Compute local level per parent (Phase 3)
- for each** parent package  $P$  with two or more direct children:
    - build sibling-only weighted graph  $G_P^{\text{loc}}$  from class refs
    - whose source and target both live in some direct child of  $P$ ;
    - Tarjan → architecture-respecting SCC break → longest-path;
    - assign `localLevel` to each direct child of  $P$ .
- 

After Step 3, a class’s architecture level is the level of the SCC node to which it belongs, measured as the longest path in the cycle-free quotient graph across all classes of the analysed project. Cycles do not influence level assignment internally because they have been collapsed before longest-path computation begins.

Step 4 treats class architecture levels and package architecture levels as independent coordinate systems. A package with no outgoing cross-package calls receives architecture level 0, regardless of the class levels inside it; two package trees with no shared dependencies are each levelled from zero in their own connected component of the weighted package graph. The earlier child–parent lift, which inflated parent package levels by the class levels their children called, is gone – it was a workaround for the missing layout coordinate.

Step 6 introduces the layout coordinate. Each parent container is treated as a self-contained scope: only class references that stay within the parent’s subtree contribute to the local sibling graph. The result is that the box for *ArchitectureView* in the *ui* package sorts above the box for the *ui.rendering* sub-package, simply because *ArchitectureView* has heavier outgoing call traffic into rendering than rendering has back into *ui*. None of this disturbs Phase 1 or Phase 2.

---

**Algorithm 2: Phase 4 – Horizontal Row Ordering (HorizontalRowLayoutOptimizer)**

---

**for each** package  $P$ :

**Step 1:** Group direct children by their `localLevel` value

**Step 2:** Collect subtree classes per row element

**Step 3:** Build weighted undirected proximity links from inter-sibling class dependencies

**Step 4:** Run bounded barycentric sweeps across rows

**Step 5:** Store the result in `horizontalLayoutOrder`

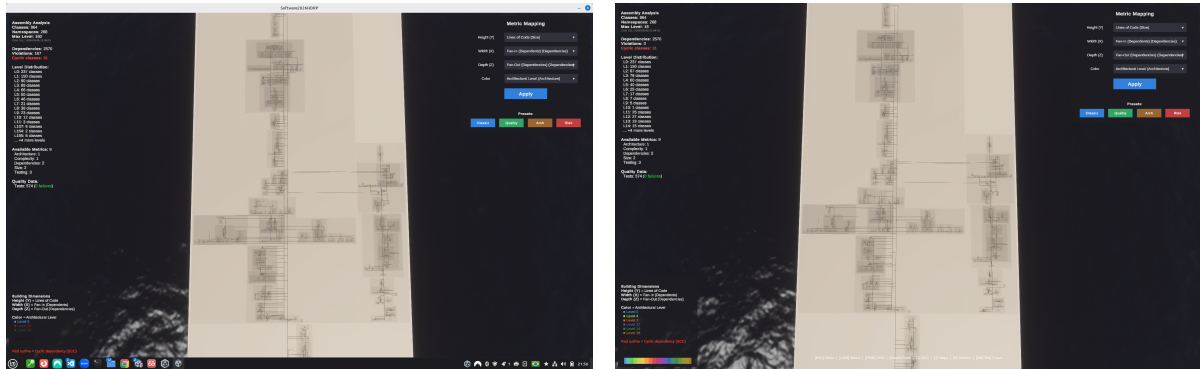
**rendering:** sort a copied child view by `localLevel` (Phase 3), then `horizontalLayoutOrder` (Phase 4), then full name

---

Phase 4 does not change any level assignment and is deliberately excluded from the layout invariants. Its only purpose is cosmetic: when an overlay of dependency arrows is displayed, same-row elements that share many inter-sibling calls are placed horizontally near each other so the arrows are shorter and less crossed. The implementation stores this as a separate `horizontalLayoutOrder` field and never mutates the semantic child list.

## R2 Fix in *software-ekg-7*: Visual Evidence

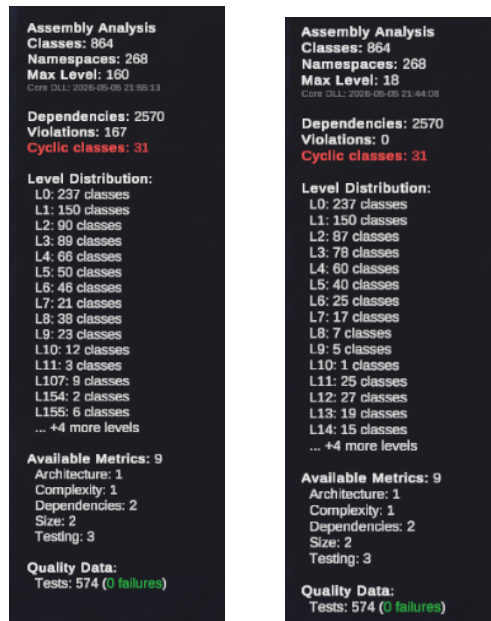
The *software-ekg-7* case is the concrete failure that motivated the package-level pipeline. Before the fix, the visualization still looked plausible: the package rectangles and dependency edges formed a readable layout, and no single visual element made the pipeline defect obvious. The metrics panel, however, exposed the inconsistency: 167 dependency violations were reported and the maximum architectural level had inflated to 160. After the package-level equalization was corrected, the same input produced zero violations and a maximum level of 18. The number of classes, packages, dependencies, cyclic classes, and tests stayed constant; the change is therefore attributable to level propagation, not to a different input model.



(a) Before the fix: the full visualization looks plausible, but the metric panel reports 167 violations and MaxLevel 160.

(b) After the fix: the same analyzed system reports zero violations and MaxLevel 18.

Figure 2: Full-screen before/after screenshots of the *software-ekg-7* visualization. The visible structure changes only subtly; the level metrics reveal the correction.



(a) Before: 167 violations, MaxLevel 160.

(b) After: zero violations, MaxLevel 18.

Figure 3: Cropped metric panels from Figure 2. These panels are the clearest user-visible symptom of the propagation defect and its fix.

## Summary

Correct level computation for layered architecture visualizations requires keeping three questions separate: where does a class sit in the global class dependency stack, where does a package sit in the weighted package dependency order, and where should an element’s box go inside its parent container? Every straightforward single-pass approach over a unified graph collapses these questions into one coordinate, and the failure modes documented above show why each obvious simplification breaks down in practice.

S202 resolves this by computing three values on three independent graphs. *Class architecture levels* are the longest path in the SCC-collapsed class dependency DAG; they represent global architectural depth. *Package architecture levels* derive from a weighted inter-package dependency graph whose edges carry aggregated method-call counts; they represent module dependency position independently of class levels. *Local levels* come from the per-parent sibling-only graph closed within that parent’s subtree; they decide visual Y position inside each box without disturbing either architecture coordinate. All three use the same algorithmic shape: Tarjan SCC detection, weight-based back-edge identification, DAG condensation, longest-path propagation.

A horizontal row-ordering pass (Phase 4) improves the legibility of dependency overlays without modifying any computed value or affecting any layout invariant. Five machine-checkable consistency checks (R1-algo, R1-visual, R2, R3, R5) run after each analysis and emit reproducer blocks for each finding. They provide regression protection and diagnostic evidence; the algorithmic contribution is the separation of the three values themselves.

Two cases demonstrate the value of the separation. In *software-ekg-7*, a plausible layered layout reported 167 dependency violations and a maximum architectural level of 160 before a package-level equalization fix; after the fix, the same input produced zero violations and a maximum level of 18 with no change to the analysed source code. On Minecraft Forge 1.19.2, introducing the local level as a distinct layout coordinate, lifting the parent filter from the weighted package graph, and preserving the computed package order in the UI instead of reordering it to suppress cycles, reduced the visual upward-edge count from roughly 6 000 to 4 200; the eliminated edges were artifacts of forcing parent packages below their sub-packages to keep them out of the algorithm’s tangle list, and the 4 200 remaining edges are real architecture violations or explicit heuristic cuts that deserve a code-level fix.

## Appendix B – LevelCalculator Reference Implementation

The following source excerpt is the complete Java implementation of the level pipeline discussed above.

```
1 package de.weigend.s202.domain.architecture;
2
3 import de.weigend.s202.domain.DomainModel;
4 import de.weigend.s202.domain.strategy.LevelCalculationStrategyContext;
5 import de.weigend.s202.graph.StronglyConnectedComponent;
6 import de.weigend.s202.graph.TarjanSCCFinder;
7 import de.weigend.s202.reader.DependencyModel;
8
9 import java.util.*;
10 import java.util.logging.Logger;
11
12 /**
13  * Calculates architectural levels for classes and packages based on dependencies.
14  *
15  * Pipeline:
16  *   Step 1 Create class objects          (all levels = 0)
17  *   Step 2 Create package objects        (all levels = 0)
18  *   Step 3 Compute class levels          strategy (Tarjan → SCC-break → DAG → longest-path)
19  *   Step 4 Compute package levels        weighted inter-package graph → SCC-break → DAG → longest-path
20  *   Step 5 Set reverse dependencies
21  *   Step 6 Assign local layer index      per-parent sibling graph → SCC-break → DAG → longest-path
22  *                                       (rendering position within each parent box, no global meaning)
23  *
24  * Package levels are computed independently from class levels using a weighted
25  * inter-package dependency graph. The weight of an edge  $P_A \rightarrow P_B$  is the
26  * aggregated method-call count from classes in  $P_A$ 's subtree to classes in  $P_B$ ,
27  * with a fallback weight of 1 for structural references without method-call data.
28  * This separates the containment hierarchy from the dependency hierarchy and
29  * correctly handles disconnected package trees (they are levelled independently,
30  * both starting at 0).
31  *
32  * Package SCC-breaking uses a weight-based rank score:
33  *    $\text{rank}(P) = (\sum \text{outgoing weights within SCC} - \sum \text{incoming weights within SCC})$ 
34  *    $\quad \quad \quad / \max(1, \text{sum of both})$ 
35  * Edge  $P_A \rightarrow P_B$  is a back-edge when  $\text{rank}(P_A) < \text{rank}(P_B) - 0.1$ .
36  * Equal-rank pairs (symmetric dependency) are left in the same SCC and assigned
37  * the same level -- they are genuine peers with no dominant dependency direction.
38  */
39 public class LevelCalculator {
40
41     private static final Logger LOG = Logger.getLogger(LevelCalculator.class.getName());
42     private static final double RANK_THRESHOLD = 0.1;
43
44     private final LevelCalculationStrategyContext strategyContext;
45
46     public LevelCalculator() {
47         this(LevelCalculationStrategyFactory.createDefault());
48     }
49
50     public LevelCalculator(LevelCalculationStrategyContext strategyContext) {
51         Objects.requireNonNull(strategyContext, "strategyContext cannot be null");
52         this.strategyContext = strategyContext;
53     }
54
55     public DomainModel calculate(DependencyModel rawModel) {
56         DomainModel model = new DomainModel();
57
58         // Step 1: Create class objects (all levels = 0)
59         for (String className : rawModel.getAllClassNames()) {
```

```

60     DependencyModel.ClassInfo rawClass = rawModel.getClass(className);
61     model.addClass(className, new DomainModel.CalculatedElementInfo(
62         className, rawClass.simpleName, "CLASS", 0,
63         new HashSet<>(rawClass.dependencies), rawClass.interfaceType));
64 }
65
66 // Step 2: Create package objects (all levels = 0)
67 for (String packageName : rawModel.getAllPackageNames()) {
68     DependencyModel.PackageInfo rawPkg = rawModel.getPackage(packageName);
69     model.addPackage(packageName, new DomainModel.CalculatedElementInfo(
70         packageName, rawPkg.simpleName, "PACKAGE", 0, new HashSet<>()));
71 }
72
73 // Step 3: Compute class levels
74 calculateClassLevels(model);
75
76 // Step 4: Compute package levels from weighted inter-package graph.
77 calculatePackageLevels(model, rawModel);
78
79 // Step 5: Set reverse dependencies
80 updateDependentRelationships(model);
81
82 // Step 6: Assign per-parent local layer index -- independent of the
83 // global architectureLevel, used by the renderer to position
84 // siblings within each parent's box.
85 new LocalLevelCalculator().assign(model, rawModel);
86
87 return model;
88 }
89
90 public LevelCalculationStrategyContext getStrategyContext() {
91     return strategyContext;
92 }
93
94 // -----
95 // Step 3 -- class levels
96 // -----
97
98 private void calculateClassLevels(DomainModel model) {
99     Map<String, Set<String>> classDeps = new HashMap<>();
100     for (DomainModel.CalculatedElementInfo c : model.getAllClasses().values()) {
101         classDeps.put(c.fullName, new HashSet<>(c.dependencies));
102     }
103     Map<String, Integer> levels =
104         strategyContext.getClassLevelStrategy().calculateClassLevels(classDeps);
105     for (Map.Entry<String, Integer> e : levels.entrySet()) {
106         DomainModel.CalculatedElementInfo info = model.getClass(e.getKey());
107         if (info != null) info.setArchitectureLevel(e.getValue());
108     }
109 }
110
111 // -----
112 // Step 4 -- package levels via weighted inter-package graph
113 // -----
114
115 private void calculatePackageLevels(DomainModel model, DependencyModel rawModel) {
116     if (model.getAllPackages().isEmpty()) return;
117     Set<String> allPkgNames = model.getAllPackages().keySet();
118
119     // Build weighted graph: weight[from][to] = aggregated method-call
120     // count from classes in 'from' to classes in 'to', with fallback
121     // weight 1 for structural dependencies without method-call data.

```

```

122 Map<String, Map<String, Integer>> weights = buildWeightedPackageGraph(model, rawModel);
123
124 // Unweighted adjacency for Tarjan (direction matters, zero-weight edges excluded)
125 Map<String, Set<String>> graph = new LinkedHashMap<>();
126 for (String pkg : allPkgNames) graph.put(pkg, new LinkedHashSet<>());
127 for (Map.Entry<String, Map<String, Integer>> entry : weights.entrySet()) {
128     graph.get(entry.getKey()).addAll(entry.getValue().keySet());
129 }
130
131 // Iteratively break SCCs using weight-based rank, then assign levels
132 // via SCC-collapsed DAG longest-path. Remaining SCCs (equal-rank peers)
133 // are collapsed to the same level without cutting any edge.
134 Set<String> backEdgeKeys = new LinkedHashSet<>(); // "from\to" -- tracked for R3
135 boolean changed = true;
136 while (changed) {
137     changed = false;
138     for (StronglyConnectedComponent scc : new TarjanSCCFinder(graph).findSCCs()) {
139         if (scc.getSize() < 2) continue;
140         Set<String> members = scc.getMembers();
141
142         // Compute weight-based rank for each member within this SCC
143         Map<String, Double> rank = new HashMap<>();
144         for (String m : members) {
145             int out = 0, in = 0;
146             for (String other : members) {
147                 if (other.equals(m)) continue;
148                 out += weights.getOrDefault(m, Map.of()).getOrDefault(other, 0);
149                 in += weights.getOrDefault(other, Map.of()).getOrDefault(m, 0);
150             }
151             rank.put(m, (out - in) / (double) Math.max(1, out + in));
152         }
153
154         // Identify back-edges: from→to where rank(from) < rank(to) - threshold
155         for (String from : new ArrayList<>(members)) {
156             for (String to : new ArrayList<>(graph.getOrDefault(from, Set.of()))) {
157                 if (!members.contains(to)) continue;
158                 if (rank.get(from) < rank.get(to) - RANK_THRESHOLD) {
159                     String key = from + "\0" + to;
160                     if (backEdgeKeys.add(key)) {
161                         graph.get(from).remove(to);
162                         changed = true;
163                     }
164                 }
165             }
166         }
167     }
168 }
169
170 // Assign package levels: SCC-collapsed DAG → longest-path
171 List<StronglyConnectedComponent> sccs = new TarjanSCCFinder(graph).findSCCs();
172 Map<String, StronglyConnectedComponent> pkgToScc = new HashMap<>();
173 for (StronglyConnectedComponent scc : sccs) {
174     for (String m : scc.getMembers()) pkgToScc.put(m, scc);
175 }
176
177 // SCC dependency graph (between SCC IDs)
178 Map<Integer, Set<Integer>> sccDeps = new HashMap<>();
179 for (StronglyConnectedComponent scc : sccs) {
180     sccDeps.put(scc.getId(), new HashSet<>());
181     for (String m : scc.getMembers()) {
182         for (String to : graph.getOrDefault(m, Set.of())) {
183             StronglyConnectedComponent toScc = pkgToScc.get(to);

```



```

184         if (toScc != null && toScc.getId() != scc.getId()) {
185             sccDeps.get(scc.getId()).add(toScc.getId());
186         }
187     }
188 }
189 }
190
191 // Longest-path levels on the SCC-collapsed DAG. The previous
192 // "childPkgUsedLevel" lift (which inflated a parent's level by
193 // the class levels its child packages used) is gone -- layout
194 // positioning lives on localLevel now, so architectureLevel
195 // can be a direct dependency-chain depth.
196 Map<Integer, Integer> sccLevels = new HashMap<>();
197 for (StronglyConnectedComponent scc : sccs) sccLevels.put(scc.getId(), 0);
198 boolean lvlChanged = true;
199 while (lvlChanged) {
200     lvlChanged = false;
201     for (StronglyConnectedComponent scc : sccs) {
202         int maxDep = -1;
203         for (int depId : sccDeps.getOrDefault(scc.getId(), Set.of())) {
204             maxDep = Math.max(maxDep, sccLevels.getOrDefault(depId, 0));
205         }
206         int newLevel = maxDep >= 0 ? maxDep + 1 : 0;
207         if (sccLevels.get(scc.getId()) != newLevel) {
208             sccLevels.put(scc.getId(), newLevel);
209             lvlChanged = true;
210         }
211     }
212 }
213
214 // Apply levels to all packages
215 Map<String, DomainModel.CalculatedElementInfo> pkgInfos = model.getAllPackages();
216 for (StronglyConnectedComponent scc : sccs) {
217     int level = sccLevels.get(scc.getId());
218     for (String member : scc.getMembers()) {
219         DomainModel.CalculatedElementInfo pkg = pkgInfos.get(member);
220         if (pkg != null) pkg.setArchitectureLevel(level);
221     }
222 }
223
224 // Store the weighted graph and identified back-edges in the model
225 model.setPackageEdgeWeights(weights);
226 model.setPackageBackEdges(backEdgeKeys);
227
228 // Populate package dependencies (unweighted) for reverse-dependency tracking
229 for (Map.Entry<String, Map<String, Integer>> entry : weights.entrySet()) {
230     DomainModel.CalculatedElementInfo pkg = pkgInfos.get(entry.getKey());
231     if (pkg != null) entry.getValue().keySet().forEach(pkg::addDependency);
232 }
233 }
234
235 /**
236  * Builds the weighted inter-package dependency graph.
237  * weight(P_A → P_B) = total method-call count from classes in P_A's subtree
238  * to classes in P_B. Method calls are a stronger signal than distinct-class
239  * counts; a class called hundreds of times clearly dominates one called once,
240  * making SCC-breaking direction unambiguous. Intra-subtree calls are excluded.
241  * Child-to-ancestor and parent-to-child edges are included.
242  * Each call count is propagated to all ancestor packages up to the point where
243  * the ancestor would enter the same subtree as the target.
244  */
245 private Map<String, Map<String, Integer>> buildWeightedPackageGraph(

```

```

246     DomainModel model, DependencyModel rawModel) {
247     Map<String, Map<String, Integer>> weights = new HashMap<>();
248     for (String pkg : model.getAllPackages().keySet()) weights.put(pkg, new LinkedHashMap<>());
249
250     Set<String> allPkgNames = weights.keySet();
251
252     for (DomainModel.CalculatedElementInfo cls : model.getAllClasses().values()) {
253         String leafPkg = extractPackageName(cls.fullName);
254         if (leafPkg == null || !allPkgNames.contains(leafPkg)) continue;
255
256         // Count method calls per target package using raw model data
257         Map<String, Integer> callCountPerPkg = new LinkedHashMap<>();
258         DependencyModel.ClassInfo rawCls = rawModel.getClass(cls.fullName);
259         if (rawCls != null) {
260             for (DependencyModel.MethodInfo method : rawCls.methods.values()) {
261                 for (Map.Entry<String, Integer> call : method.methodCalls.entrySet()) {
262                     // Method call key format: "ownerClass.methodName"
263                     // Find the target class by checking known dependencies
264                     String calledKey = call.getKey();
265                     for (String dep : cls.dependencies) {
266                         if (calledKey.startsWith(dep + ".")) {
267                             String toPkg = extractPackageName(dep);
268                             if (toPkg != null && !leafPkg.equals(toPkg)
269                                 && allPkgNames.contains(toPkg)) {
270                                 callCountPerPkg.merge(toPkg, call.getValue(), Integer::sum);
271                             }
272                             break;
273                         }
274                     }
275                 }
276             }
277         }
278         // Fall back to 1 per target package for structural deps with no call data
279         for (String dep : cls.dependencies) {
280             String toPkg = extractPackageName(dep);
281             if (toPkg != null && !leafPkg.equals(toPkg) && allPkgNames.contains(toPkg)) {
282                 callCountPerPkg.putIfAbsent(toPkg, 1);
283             }
284         }
285
286         // Propagate each weight up to ancestor packages. Parent->child
287         // edges are no longer filtered here -- the rank-based SCC breaker
288         // decides direction based on actual call-count weight. Layout
289         // positioning is the LocalLevelCalculator's job (step 6), so the
290         // global package level can be a direct dependency-chain count.
291         for (Map.Entry<String, Integer> entry : callCountPerPkg.entrySet()) {
292             String toPkg = entry.getKey();
293             int callCount = entry.getValue();
294             String ancestor = leafPkg;
295             while (ancestor != null && allPkgNames.contains(ancestor)) {
296                 if (ancestor.equals(toPkg)) break;
297                 weights.get(ancestor).merge(toPkg, callCount, Integer::sum);
298                 ancestor = extractPackageName(ancestor);
299             }
300         }
301     }
302     return weights;
303 }
304
305 // -----
306 // Step 5 -- reverse dependencies
307 // -----

```

```

308
309     private void updateDependentRelationships(DomainModel model) {
310         for (DomainModel.CalculatedElementInfo cls : model.getAllClasses().values()) {
311             for (String dep : cls.dependencies) {
312                 DomainModel.CalculatedElementInfo d = model.getClass(dep);
313                 if (d != null) d.addDependent(cls.fullName);
314             }
315         }
316         for (DomainModel.CalculatedElementInfo pkg : model.getAllPackages().values()) {
317             for (String dep : pkg.dependencies) {
318                 DomainModel.CalculatedElementInfo d = model.getPackage(dep);
319                 if (d != null) d.addDependent(pkg.fullName);
320             }
321         }
322     }
323
324     // -----
325     // Utilities
326     // -----
327
328     private static boolean isInSameSubtree(String a, String b) {
329         return a.startsWith(b + ".") || b.startsWith(a + ".");
330     }
331
332     private static String extractPackageName(String className) {
333         if (className == null || !className.contains(".")) return null;
334         return className.substring(0, className.lastIndexOf('.'));
335     }
336 }

```

The companion class `LocalLevelCalculator` implements Phase 3 (Step 6).

```

1 package de.weigend.s202.domain.architecture;
2
3 import de.weigend.s202.domain.DomainModel;
4 import de.weigend.s202.domain.DomainModel.CalculatedElementInfo;
5 import de.weigend.s202.graph.StronglyConnectedComponent;
6 import de.weigend.s202.graph.TarjanSCCFinder;
7 import de.weigend.s202.reader.DependencyModel;
8
9 import java.util.ArrayList;
10 import java.util.HashMap;
11 import java.util.HashSet;
12 import java.util.LinkedHashSet;
13 import java.util.List;
14 import java.util.Map;
15 import java.util.Set;
16
17 /**
18  * Assigns a per-parent layer position ({@code localLevel}) to every
19  * element in a {@link DomainModel}. Each parent package is treated as a
20  * self-contained scope: only class dependencies whose source <em>and</em>
21  * target both live in the parent's subtree contribute to the sibling-only
22  * weighted graph. Refs that leave the parent are ignored -- they belong to
23  * the global {@code architectureLevel} via the package hierarchy.
24  *
25  * <p>Within each parent's sibling graph:
26  * <ol>
27  *   <li>Tarjan finds SCCs.</li>
28  *   <li>SCCs of size > 1 are broken one edge at a time. The default
29  *       strategy evaluates every candidate edge in the whole SCC and cuts
30  *       the edge whose removal decomposes the cyclic component the most.

```

```

31 *      Package-architecture contradictions, edge weight, and rank are used
32 *      only as tie-breakers.</li>
33 *      <li>Longest-path on the SCC-collapsed DAG assigns a {@code layer}
34 *      (= local layer index) per sibling. Layer 0 = bottom of the box,
35 *      higher values = visually higher.</li>
36 * </ol>
37 *
38 * <p>Single-child containers stay at layer 0 -- no graph, no work.
39 *
40 * <p>The algorithm intentionally mirrors the structure of
41 * {@link LevelCalculator}'s package-level computation, but applied
42 * locally and using a different graph. Unlike package-level computation,
43 * local SCC breaking is layout-oriented: it aims to stabilize the visible
44 * architecture by reducing cyclic local structure, not to minimize the
45 * number of visible upward edges. Local cycles are not surfaced as tangles;
46 * the user-visible tangle list remains the global one.
47 */
48 public class LocalLevelCalculator {
49
50     private static final double RANK_THRESHOLD = 0.1;
51     private final BreakMode breakMode;
52
53     public LocalLevelCalculator() {
54         this(BreakMode.MAX_CYCLE_BREAK);
55     }
56
57     LocalLevelCalculator(BreakMode breakMode) {
58         this.breakMode = breakMode;
59     }
60
61     enum BreakMode {
62         /**
63          * Legacy local heuristic: remove every edge whose source has lower
64          * weighted rank than its target by more than {@link #RANK_THRESHOLD}.
65          */
66         RANK,
67         /**
68          * Default local heuristic: inspect the whole SCC and remove one edge
69          * whose cut maximally decomposes the cyclic component.
70          */
71         MAX_CYCLE_BREAK
72     }
73
74     public void assign(DomainModel domain, DependencyModel rawModel) {
75         Map<String, List<CalculatedElementInfo>> childrenByParent = groupChildrenByParent(domain);
76         for (Map.Entry<String, List<CalculatedElementInfo>> entry : childrenByParent.entrySet()) {
77             assignForParent(entry.getValue(), domain, rawModel);
78         }
79     }
80
81     private void assignForParent(List<CalculatedElementInfo> siblings,
82                                 DomainModel domain, DependencyModel rawModel) {
83         if (siblings.size() <= 1) {
84             return; // single child stays at layer 0
85         }
86
87         Set<String> siblingFqns = new LinkedHashSet<>();
88         for (CalculatedElementInfo c : siblings) {
89             siblingFqns.add(c.fullName);
90         }
91
92         Map<String, Map<String, Integer>> weights = buildSiblingGraph(siblingFqns, domain, rawModel);

```

```

93     Map<String, Integer> layers = computeLayers(weights, siblings, siblingFqns);
94     for (CalculatedElementInfo s : siblings) {
95         s.setLocalLayerIndex(layers.getOrDefault(s.fullName, 0));
96     }
97 }
98
99 /**
100  * Build the weighted sibling-only edge map. For every class C in any
101  * sibling's subtree, every dep into another sibling's subtree
102  * contributes {@code callCount(C, dep)} weight; deps that leave the
103  * parent's subtree or stay inside the same sibling are skipped.
104  */
105 private Map<String, Map<String, Integer>> buildSiblingGraph(Set<String> siblingFqns,
106                                                         DomainModel domain,
107                                                         DependencyModel rawModel) {
108     Map<String, Map<String, Integer>> weights = new HashMap<>();
109     for (String s : siblingFqns) {
110         weights.put(s, new HashMap<>());
111     }
112     for (CalculatedElementInfo cls : domain.getAllClasses().values()) {
113         String fromSibling = containingSibling(cls.fullName, siblingFqns);
114         if (fromSibling == null) {
115             continue;
116         }
117         for (String dep : cls.dependencies) {
118             if (domain.getClass(dep) == null) {
119                 continue; // external -- not in any sibling
120             }
121             String toSibling = containingSibling(dep, siblingFqns);
122             if (toSibling == null || toSibling.equals(fromSibling)) {
123                 continue; // out of parent OR intra-sibling
124             }
125             int w = callCount(cls.fullName, dep, rawModel);
126             weights.get(fromSibling).merge(toSibling, w, Integer::sum);
127         }
128     }
129     return weights;
130 }
131
132 /**
133  * Walk up the fqcn package chain until the first ancestor that is one
134  * of the siblings, or return {@code null} when none of the ancestors
135  * is -- the class lives outside the parent's subtree.
136  */
137 private static String containingSibling(String fqcn, Set<String> siblingFqns) {
138     String current = fqcn;
139     while (current != null && !current.isEmpty()) {
140         if (siblingFqns.contains(current)) {
141             return current;
142         }
143         int dot = current.lastIndexOf('.');
144         if (dot < 0) {
145             return null;
146         }
147         current = current.substring(0, dot);
148     }
149     return null;
150 }
151
152 private static Map<String, List<CalculatedElementInfo>> groupChildrenByParent(DomainModel domain) {
153     Map<String, List<CalculatedElementInfo>> result = new HashMap<>();
154     for (CalculatedElementInfo cls : domain.getAllClasses().values()) {

```

```

155         result.computeIfAbsent(parentOf(cls.fullName), k -> new ArrayList<>()).add(cls);
156     }
157     for (CalculatedElementInfo pkg : domain.getAllPackages().values()) {
158         result.computeIfAbsent(parentOf(pkg.fullName), k -> new ArrayList<>()).add(pkg);
159     }
160     return result;
161 }
162
163 private static String parentOf(String fqn) {
164     int dot = fqn.lastIndexOf('.');
165     return dot < 0 ? "" : fqn.substring(0, dot);
166 }
167
168 /**
169  * Method-call count from class {@code from} to class {@code to}.
170  * Falls back to 1 for structural deps (extends/implements/field type)
171  * where no method-call info exists -- same convention the global
172  * weighted package graph uses.
173  */
174 private static int callCount(String from, String to, DependencyModel rawModel) {
175     DependencyModel.ClassInfo cls = rawModel.getClass(from);
176     if (cls == null) {
177         return 1;
178     }
179     int count = 0;
180     String prefix = to + ".";
181     for (DependencyModel.MethodInfo method : cls.methods.values()) {
182         for (Map.Entry<String, Integer> call : method.methodCalls.entrySet()) {
183             if (call.getKey().startsWith(prefix)) {
184                 count += call.getValue();
185             }
186         }
187     }
188     return count > 0 ? count : 1;
189 }
190
191 // ----- SCC-collapsed longest path with configurable SCC break -----
192
193 private Map<String, Integer> computeLayers(Map<String, Map<String, Integer>> weights,
194                                           List<CalculatedElementInfo> siblings,
195                                           Set<String> nodes) {
196     Map<String, Set<String>> graph = new HashMap<>();
197     for (String n : nodes) {
198         graph.put(n, new HashSet<>(weights.getOrDefault(n, Map.of()).keySet()));
199     }
200     Map<String, CalculatedElementInfo> siblingByName = new HashMap<>();
201     for (CalculatedElementInfo sibling : siblings) {
202         siblingByName.put(sibling.fullName, sibling);
203     }
204
205     // Iteratively break local SCCs, then assign layers on the remaining DAG.
206     // MAX_CYCLE_BREAK is the default; RANK remains available as a legacy
207     // comparison mode for tests and experiments.
208     boolean changed = true;
209     while (changed) {
210         changed = false;
211         for (StronglyConnectedComponent scc : new TarjanSCCFinder(graph).findSCCs()) {
212             if (scc.getSize() < 2) {
213                 continue;
214             }
215             Set<String> members = scc.getMembers();
216             Map<String, Double> rank = new HashMap<>();

```

```

217     for (String m : members) {
218         int out = 0;
219         int in = 0;
220         for (String other : members) {
221             if (other.equals(m)) {
222                 continue;
223             }
224             out += weights.getOrDefault(m, Map.of()).getOrDefault(other, 0);
225             in += weights.getOrDefault(other, Map.of()).getOrDefault(m, 0);
226         }
227         rank.put(m, (out - in) / (double) Math.max(1, out + in));
228     }
229     if (breakMode == BreakMode.MAX_CYCLE_BREAK) {
230         EdgeCutCandidate best = bestCycleBreakingEdge(graph, weights, members, rank, siblingByName);
231         if (best != null) {
232             graph.get(best.from()).remove(best.to());
233             changed = true;
234         }
235         continue;
236     }
237     // Legacy path: remove all rank-based back-edges found in this SCC.
238     for (String from : new ArrayList<>(members)) {
239         for (String to : new ArrayList<>(graph.getOrDefault(from, Set.of()))) {
240             if (!members.contains(to)) {
241                 continue;
242             }
243             if (rank.get(from) < rank.get(to) - RANK_THRESHOLD) {
244                 graph.get(from).remove(to);
245                 changed = true;
246             }
247         }
248     }
249 }
250
251
252 // SCC-collapsed longest-path level assignment.
253 List<StronglyConnectedComponent> sccs = new TarjanSCCFinder(graph).findSCCs();
254 Map<String, StronglyConnectedComponent> nodeToScc = new HashMap<>();
255 for (StronglyConnectedComponent scc : sccs) {
256     for (String m : scc.getMembers()) {
257         nodeToScc.put(m, scc);
258     }
259 }
260 Map<Integer, Set<Integer>> sccDeps = new HashMap<>();
261 for (StronglyConnectedComponent scc : sccs) {
262     sccDeps.put(scc.getId(), new HashSet<>());
263     for (String m : scc.getMembers()) {
264         for (String to : graph.getOrDefault(m, Set.of())) {
265             StronglyConnectedComponent toScc = nodeToScc.get(to);
266             if (toScc != null && toScc.getId() != scc.getId()) {
267                 sccDeps.get(scc.getId()).add(toScc.getId());
268             }
269         }
270     }
271 }
272 Map<Integer, Integer> sccLayers = new HashMap<>();
273 for (StronglyConnectedComponent scc : sccs) {
274     sccLayers.put(scc.getId(), 0);
275 }
276 boolean lvlChanged = true;
277 while (lvlChanged) {
278     lvlChanged = false;

```

```

279     for (StronglyConnectedComponent scc : sccs) {
280         int maxDep = -1;
281         for (int depId : sccDeps.get(scc.getId())) {
282             maxDep = Math.max(maxDep, sccLayers.get(depId));
283         }
284         int newLayer = maxDep >= 0 ? maxDep + 1 : 0;
285         if (sccLayers.get(scc.getId()) != newLayer) {
286             sccLayers.put(scc.getId(), newLayer);
287             lvlChanged = true;
288         }
289     }
290 }
291
292 Map<String, Integer> result = new HashMap<>();
293 for (StronglyConnectedComponent scc : sccs) {
294     int layer = sccLayers.get(scc.getId());
295     for (String m : scc.getMembers()) {
296         result.put(m, layer);
297     }
298 }
299 return result;
300 }
301
302 private static EdgeCutCandidate bestCycleBreakingEdge(Map<String, Set<String>> graph,
303                                                       Map<String, Map<String, Integer>> weights,
304                                                       Set<String> members,
305                                                       Map<String, Double> rank,
306                                                       Map<String, CalculatedElementInfo> siblingByName) {
307     EdgeCutCandidate best = null;
308     for (String from : members) {
309         for (String to : graph.getOrDefault(from, Set.of())) {
310             if (!members.contains(to)) {
311                 continue;
312             }
313             EdgeCutCandidate candidate = new EdgeCutCandidate(
314                 from,
315                 to,
316                 cycleBreakScore(graph, members, from, to),
317                 architectureContradiction(from, to, siblingByName),
318                 weights.getOrDefault(from, Map.of()).getOrDefault(to, 0),
319                 rank.getOrDefault(to, 0.0) - rank.getOrDefault(from, 0.0));
320             if (best == null || candidate.compareTo(best) > 0) {
321                 best = candidate;
322             }
323         }
324     }
325     return best;
326 }
327
328 /**
329  * Score how much one candidate cut decomposes the current SCC. The score
330  * compares the original SCC size squared with the sum of squared SCC sizes
331  * after removing the candidate edge. A cut that splits one large cycle into
332  * several smaller components therefore wins over a cut that leaves the SCC
333  * almost intact.
334  */
335 private static int cycleBreakScore(Map<String, Set<String>> graph,
336                                   Set<String> members,
337                                   String cutFrom,
338                                   String cutTo) {
339     Map<String, Set<String>> copy = new HashMap<>();
340     for (String member : members) {

```



```

341         copy.put(member, new HashSet<>());
342         for (String to : graph.getDefault(member, Set.of())) {
343             if (members.contains(to) && !(member.equals(cutFrom) && to.equals(cutTo))) {
344                 copy.get(member).add(to);
345             }
346         }
347     }
348
349     int after = 0;
350     for (StronglyConnectedComponent scc : new TarjanSCCFinder(copy).findSCCs()) {
351         after += scc.getSize() * scc.getSize();
352     }
353     return members.size() * members.size() - after;
354 }
355
356 /**
357  * Tie-breaker for package siblings: if an edge runs against the already
358  * computed package architectureLevel, prefer cutting it over an otherwise
359  * equivalent package edge. Class edges return 0 because classes are placed
360  * inside the package frame, not used to redefine that frame.
361  */
362 private static int architectureContradiction(String from,
363                                             String to,
364                                             Map<String, CalculatedElementInfo> siblingByName) {
365     CalculatedElementInfo fromInfo = siblingByName.get(from);
366     CalculatedElementInfo toInfo = siblingByName.get(to);
367     if (fromInfo == null || toInfo == null) {
368         return 0;
369     }
370     if (!"PACKAGE".equals(fromInfo.type) || !"PACKAGE".equals(toInfo.type)) {
371         return 0;
372     }
373     return fromInfo.architectureLevel < toInfo.architectureLevel ? 1 : 0;
374 }
375
376 private record EdgeCutCandidate(String from,
377                                String to,
378                                int cycleBreakScore,
379                                int architectureContradiction,
380                                int weight,
381                                double rankBackEdgeScore)
382     implements Comparable<EdgeCutCandidate> {
383     /**
384      * Higher is better. Order of preference:
385      * 1. always cut an edge that contradicts the package architecture
386      *    hypothesis first -- Phase 3 must never reorder packages relative
387      *    to what Phase 2 computed,
388      * 2. among architecture-equivalent candidates, decompose the local SCC
389      *    as much as possible,
390      * 3. cut the weaker edge,
391      * 4. fall back to the legacy rank signal.
392      */
393     @Override
394     public int compareTo(EdgeCutCandidate other) {
395         int cmp = Integer.compare(architectureContradiction, other.architectureContradiction);
396         if (cmp != 0) {
397             return cmp;
398         }
399         cmp = Integer.compare(cycleBreakScore, other.cycleBreakScore);
400         if (cmp != 0) {
401             return cmp;
402         }

```

```
403         cmp = Integer.compare(other.weight, weight);
404         if (cmp != 0) {
405             return cmp;
406         }
407         return Double.compare(rankBackEdgeScore, other.rankBackEdgeScore);
408     }
409 }
410 }
```

## Appendix C – Questions and Answers

---

### **Why does S202 need a separate hierarchy for packages?**

Because S202 orders classes and packages by different criteria. A class level is the depth of a node in the class dependency DAG after SCC handling: if class  $A$  depends on class  $B$ , then  $A$  belongs above  $B$  unless the edge is an explicit heuristic back-edge or both classes remain in the same SCC.

A package level answers a different question. Packages are containers, and a single package can contain many classes with dependencies in both directions. S202 therefore cannot derive package order from the deepest class inside the package. Instead, it compares the weighted relations between packages: when the traffic from package  $P_A$  to package  $P_B$  dominates the reverse traffic,  $P_A$  becomes the higher dependent package and  $P_B$  the lower package it depends on. S202 keeps equal or near-equal bidirectional traffic as a cyclic peer relationship.

This is why S202 computes class architecture levels and package architecture levels on separate graphs. The separation prevents class depth from accidentally lifting containers and prevents package aggregation from changing the meaning of class-level dependency depth.

It also prevents a feedback loop. If S202 derived package levels from class levels and then fed them back into class or layout ordering, each update could change the input of the next update. A package moves because one contained class moves; that package movement then changes sibling order or parent order; the changed order can expose another cycle or another dominant relation. Sorting and leveling algorithms then no longer operate on a stable graph. The result can depend on traversal order, classpath order, or on how often the loop is run. The system may converge accidentally, oscillate, or settle on a layout that is only an artifact of the update sequence. Separating the graphs makes each phase operate on fixed input and gives the invariants a well-defined result to check.

### **Why does S202 need both an architecture level and a local UI level?**

Because these two numbers answer different questions. The architecture level of a class says how deep the class is in the global dependency graph: how many dependency steps separate it from the lower end of the class DAG after SCC handling. This is a system-wide analysis value.

The local level answers a layout question: where should this class or sub-package be placed relative to the other direct children of the same package? That question cannot be answered reliably from the global architecture level alone. A class can have a high global architecture level because it depends on deep chains elsewhere in the system, even though inside its own package it is not the highest local element. Conversely, a package-level coordinator may need to sit above a sub-package it calls into, even if their global architecture levels do not express that local relationship.

Using only the global DAG level for layout therefore over-constrains the picture. Elements can be pulled too high by dependencies outside their local container, and subsequent sorting passes try to repair a layout decision with the wrong input value. That is hard to make stable because the global level and the local sibling order are coupled through unrelated dependencies.

S202 keeps both values. The architecture level remains the semantic dependency depth used for global checks and reports. The local level is recomputed inside each package from the sibling-only graph and is used for visual ordering inside that package. The renderer can then place elements correctly within a container without changing what their global architecture level means.